The Pyramid Puzzle

Whodunnit? · Triangles & Congruence Name: _____

At the foot of an ancient pyramid, a team of archaeologists was ready to unveil their greatest discovery: the **Golden Scarab**, a jeweled beetle 3,000 years old. But when the cloth was pulled away, the display stand was empty!

Six people were at the dig site, each in a different place, and there are six clever ways to carry a scarab away. You are the expedition detective: find **who** took it, **where** they were, and **how** they got it out.

How to crack the case

1. Your teacher will post **10 clue cards** around the room. Each card has a math problem and four answer choices. Every choice is paired with a clue.
2. Solve the problem on the **Show Your Work** page. Find your answer among the choices and read the clue next to it. If your answer is not there, check your work.
3. Use the clue to cross out suspects, places or methods on the case board below. Write the card number beside what it cleared.
4. After all 10 cards, exactly one suspect, one place and one method are left. Those three boxes name the culprit. If you end up with none, or more than one, a clue went wrong: recheck your math!

The Suspects	The Places	The Methods
☐ Dr. Amara Lead Archaeologist	☐ Pyramid Entrance	☐ Hidden in a saddlebag
☐ Mr. Petrie Museum Rep	☐ Burial Chamber	☐ Buried in the sand
☐ Zed Tour Guide	☐ Dig Site	☐ Wrapped in mummy cloth
☐ Ms. Okonkwo Photographer	☐ Camp Tent	☐ Disguised as a souvenir
☐ Prof. Lyle Hieroglyph Expert	☐ Gift Stall	☐ Packed in a camera case
☐ Hassan Camel Driver	☐ Oasis	☐ Poured into a water jug

The culprit is _____ **in** _____ **using** _____

Show Your Work

The Pyramid Puzzle · Show every step in the matching box. Name: _____

1. Choice _____	**2.** Choice _____
3. Choice _____	**4.** Choice _____
5. Choice _____	**6.** Choice _____
7. Choice _____	**8.** Choice _____
9. Choice _____	**10.** Choice _____

Clue #1

The Pyramid Puzzle

In △ABC and △DEF: AB ≅ DE, BC ≅ EF and AC ≅ DF. Which congruence rule proves △ABC ≅ △DEF?

A. HL	The water jugs were all poured out at noon. Only plain water ran out.
B. SSS	Dr. Amara was brushing sand from a vase at the Dig Site. Her whole team saw her.
C. ASA	Dr. Amara was measuring the walls of the Burial Chamber. Her tape measure is still on the floor.
D. SAS	Ms. Okonkwo was photographing the carvings at the Pyramid Entrance. Her flash lit up the doorway.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #2

The Pyramid Puzzle

In △ABC and △DEF: AB ≅ DE, ∠B ≅ ∠E and BC ≅ EF. Which rule proves the triangles congruent?

A. SAS	The camel saddlebags were inspected. Only dates and bread inside.
B. SSS	Ms. Okonkwo was snapping pictures of the diggers at the Dig Site. The shots are time-stamped.
C. AAS	The big clay jug is cracked and leaks. Nothing could be stored in it.
D. ASA	Zed was telling spooky stories in the Burial Chamber. His whole tour group was there.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #3

The Pyramid Puzzle

In △ABC and △DEF: ∠A ≅ ∠D, AB ≅ DE and ∠B ≅ ∠E. Which rule proves the triangles congruent?

A. SAS	Ms. Okonkwo was changing film in the Camp Tent. The flap was tied shut with her inside.
B. SSS	Every water jug was filled from the oasis well. All of them hold only water.
C. ASA	Mr. Petrie was filling out museum forms in the Camp Tent. The forms are signed and timed.
D. AAS	Prof. Lyle was copying hieroglyphs in the Burial Chamber. His notebook has the proof.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #4

The Pyramid Puzzle

In △ABC and △DEF: ∠A ≅ ∠D, ∠B ≅ ∠E and BC ≅ EF. Which rule proves the triangles congruent?

A. HL	The water jug has a very narrow mouth. Nothing bigger than a pebble could be poured in.
B. AAS	The sand around the camp was raked and sifted. Nothing was buried.
C. ASA	Ms. Okonkwo was bargaining at the Gift Stall for a postcard. The shopkeeper remembers her.
D. SAS	Hassan was carrying water into the Burial Chamber for the workers. Wet footprints trail behind him.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #5

The Pyramid Puzzle

Two right triangles have congruent hypotenuses and one pair of congruent legs. Which rule proves them congruent?

A. HL	Zed was guiding a tour group at the Pyramid Entrance. Twenty tourists remember him.
B. SSS	Mr. Petrie was inspecting the Burial Chamber with a clipboard. The guard signed him in.
C. SAS	Ms. Okonkwo was taking sunset photos at the Oasis. The palm trees are in every shot.
D. Not enough info	The jugs were weighed at the gate. Every one weighed exactly what water should.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

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Clue #6

The Pyramid Puzzle

In △ABC and △DEF: AB ≅ DE, BC ≅ EF and ∠A ≅ ∠D. What can you conclude?

A. SAS	Ms. Okonkwo was adjusting her lens at the Pyramid Entrance. A tourist asked her for directions.
B. ASA	A jug was tipped over by the camp. The spilled water was clear, with nothing in it.
C. SSS	Zed was pointing out wall paintings in the Burial Chamber. The tour group clapped.
D. Not enough info	The mummy cloths are all old and dusty. None was recently unwrapped.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #7

The Pyramid Puzzle

△ABC ≅ △DEF. m∠B = (4x + 8)° and m∠E = 56°. Find x.

A. x = 29	The water jug is sealed with wax, and the wax is old and unbroken.
B. x = 14	Dr. Amara was dusting off a statue inside the Burial Chamber. Her assistant held the lamp.
C. x = 12	Prof. Lyle was buying a scarab keychain at the Gift Stall. The receipt has the time.
D. x = 16	Ms. Okonkwo was photographing a sandstorm rolling across the Dig Site. The diggers took cover beside her.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #8

The Pyramid Puzzle

△ABC ≅ △DEF. AB = 3x − 2 and DE = 2x + 7. Find x.

A. x = 7	The camp’s jugs were measured and found far too tiny to hold a jeweled beetle.
B. x = 9	The souvenir stalls are filled with cheap plastic beetles. The real Scarab isn’t among them.
C. x = −5	Prof. Lyle was translating a wall inscription in the Burial Chamber. The guard watched him.
D. x = 5	Ms. Okonkwo was charging camera batteries in the Camp Tent. The generator hummed all night.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #9

The Pyramid Puzzle

△PQR ≅ △XYZ. Which angle corresponds to ∠Q?

A. ∠Z	Hassan was delivering lamp oil to the Burial Chamber. The workers signed for it.
B. ∠X	Ms. Okonkwo was buying film at the Gift Stall. The receipt shows the time.
C. ∠R	The camp’s water jugs were rinsed and left upside down to dry. None was used last night.
D. ∠Y	Hassan was giving the camels water at the Oasis. A farmer saw him there.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #10

The Pyramid Puzzle

Two triangles share the side AC. Which property justifies AC ≅ AC?

A. Reflexive	The camera case was opened at the gate. Just lenses and batteries.
B. Symmetric	Ms. Okonkwo was watching camels drink at the Oasis. She took a photo of the whole herd.
C. Substitution	The jug on the supply cart has a hole in the bottom. It could not hold anything.
D. Transitive	Mr. Petrie was counting display cases in the Burial Chamber. The tally sheet is dated.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.